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Genotypic Profiling, Functional Analysis, Cholesterol-Lowering Ability, and Angiotensin I-Converting Enzyme (ACE) Inhibitory Activity of Probiotic *Lactiplantibacillus plantarum* K25 via Different Approaches

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Abstract

Due to its alleged health advantages, several uses in biotechnology and food safety, the well-known probiotic strain *Lactiplantibacillus plantarum* K25 has drawn interest. This in-depth investigation explores the genetic diversity, makeup, and security characteristics of the microbial genome of *L. plantarum* K25, providing insightful knowledge about its genotypic profile and functional characteristics. Utilizing cutting-edge bioinformatics techniques like comparative genomics, pan-genomics, and genotypic profiling was carried out to reveal the strain's multidimensional potential in various fields. The results not only add to our understanding of the genetic makeup of *L. plantarum* K25 but also show off its acceptability in various fields, notably in biotechnology and food safety. The explanation of evolutionary links, which highlights *L. plantarum* K25's aptitude as a probiotic, is one notable finding from this research. Its safety profile, which is emphasized by the absence of genes linked to antibiotic resistance, is crucial and supports its status as a promising probiotic option.

Keywords Probiotic · *Lactiplantibacillus* · Pan genomics · Comparative genomics · Genotyping

Introduction

Understanding the genetic variations of probiotic microbes has become critical in an era of growing microbiome research and customized treatment. *Lactiplantibacillus plantarum* K25, a strain known for its potential probiotic characteristics, is an intriguing subject of inquiry [1]. This work dives into the genetic diversity within the genome of *L. plantarum* K25, revealing information on its genotypic profile, and functional features. The ramifications of this extensive research extend to both the probiotic industry and safety initiatives, as a detailed understanding of the strain's genetic makeup holds the key to properly exploiting its probiotic potential while assuring the highest safety standards [2]. The fast growth of human microbiome research has revealed the tremendous influence of probiotic microbes on human health and well-being. Among them, *Lactiplantibacillus plantarum* strains have received attention for their potential to provide a variety of health advantages ranging from improved gut health to immune system regulation [1]. However, fully exploiting their probiotic potential requires a thorough understanding of their genetic roots. Furthermore, deciphering the genotypic profile of *L. plantarum* K25 might

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give useful information regarding its physiological characteristics and prospective functional uses [3]. Furthermore, as the probiotic market grows, ensuring the safety of these microorganisms becomes increasingly important, making a thorough investigation into the genetic variation and functional attributes of *L. plantarum* K25 essential for guiding its responsible and effective use in probiotic products and therapeutic interventions [4].

Despite the known promise of *L. plantarum* K25 as a probiotic candidate, we still have a lot to learn about its genetic diversity, genotypic profile, and functional characteristics [5]. This lack of information impedes the full realization of its probiotic potential and safe deployment, limiting its practical uses in personalized healthcare and preventative medicine. Closing this gap via a thorough examination is critical for reaping the benefits of *L. plantarum* K25 while properly addressing safety issues [5]. The goal of this study is to gain a comprehensive understanding of *L. plantarum* K25 through whole-genome sequencing, conducting genotypic profiling to uncover physiological traits, and performing functional analyses to assess its potential applications as a probiotic. Furthermore, the study aims to evaluate safety concerns by finding any relevant genetic variables linked to virulence. Finally, the study hopes to close the information gap around *L. plantarum* K25, allowing for its optimal use as a safe and effective probiotic strain in a variety of health-promoting scenarios. This study contributes to the existing probiotics literature by investigating their potential impact on gut health and immune function in a varied population. Its goal is to gain a better understanding of how these microorganisms contribute to overall well-being by studying the impacts of a single strain. Such studies improve our understanding of probiotics' significance in human health and inform prospective medicinal uses [6].

In conclusion, this thorough study of *L. plantarum* K25 genomic diversity, genotypic profiling, and functional features has offered useful insights into its probiotic potential and safety issues. The discovery of its genetic makeup, adaptive processes, and physiological features opens the door to individualized treatment and preventative medicine [3]. The findings emphasize the significance of using this strain responsibly and add to the expanding body of information targeted at improving probiotic effectiveness while maintaining the highest safety requirements. *Lactiplantibacillus plantarum* strains are resilient in the gut environment, supporting digestive health by altering gut microbiota and improving barrier function. They provide immunological support through immunomodulatory actions and may include antioxidants, adding to general health. As probiotics emerge as potential agents for boosting human health, the findings of this study lay the groundwork for future

research and practical application of *L. plantarum* K25 in a variety of therapeutic and nutritional scenarios [7].

Materials and Methods

Bacterial Strain, Culture Conditions, and EPS Extraction

Lactiplantibacillus plantarum K25 (*L. plantarum* K25) was obtained from Tibetan Kefir grains in 2015 and preserved as frozen stocks at 9 °C in MRS broth enriched with 20% (v/v) glycerol. The identification of these strains involved employing gram staining, catalase tests, and an assessment of cellular morphology. The API 50 CHL test from bioMérieux in Marcy-l'Étoile, France, was utilized for strain characterization, and 16S rDNA sequencing was conducted following established protocols [3, 8–14]. For obtaining the EPS, the method of Jiang et al. [15] was followed. The polysaccharide *L. plantarum* K25 was grown at 37 °C for 24 h in SDM and added with a selected concentration of CaCl_2 or without the addition of CaCl_2 (control). The EPS was extracted and purified following the method described by Wang et al. [12].

DNA Extraction and Whole-Genome Sequencing

The genomic DNA extraction process involved the use of Promega's Wizard® Genomic DNA Purification Kit. Subsequently, the DNA was quantified using a TBS-380 fluorometer from Turner Bio Systems Inc. in Sunnyvale, CA, USA, with an insert size of 15 Kb. DNA samples meeting high-quality criteria, defined as having an OD_{260/280} ratio ranging from 1.8 to 2.0 and a quantity exceeding 20 µg, were selected for further analysis.

For the construction of Illumina sequencing libraries, sheared DNA fragments were prepared using the NEX-Tflex™ Rapid DNA-Seq Kit. In brief, the 5' ends of the fragments underwent initial end repair and phosphorylation, followed by A-tailing and ligation with sequencing adapters at the 3' termini. The final step involved PCR amplification to enrich the adapter-ligated products. These libraries were then subjected to paired-end Illumina sequencing on an Illumina HiSeq X Ten platform, producing reads with a length of 150 bp for each end.

The genome of *L. plantarum* K25 underwent sequencing utilizing both single molecule real-time (SMRT) technology and the Illumina sequencing platform, resulting in the accession number GCA_003020005.1. The Illumina data was employed to estimate the genome's complexity.

Genomic DNA was extracted using the Qiagen DNA extraction kit, following the manufacturer's instructions, during the isolation period spanning from March to May of 2019 [16].

Angiotensin I-Converting Enzyme (ACE) Inhibitory Activity of K25 EPS

The ACE inhibition activity was evaluated by the o-phthalaldehyde colorimetric method with slight modifications [17]. The ACE solution, with a concentration of 25 mU/mL, was prepared in a 0.1 mol/L borate buffer at pH 8.3, supplemented with 0.3 mol/L NaCl. Simultaneously, the HHL (hippuryl-L-histidyl-L-leucine) solution, at a concentration of 5 mmol/L, was created in a 0.1 mol/L phosphate buffer at pH 7, containing 0.3 mol/L NaCl.

In the initial step, 200 µL of the K25 EPS solution, at various concentrations (0.5, 1, 2, and 3 mg/mL), was combined with 80 µL of the ACE solution and incubated at 37 °C for 5 min. Subsequently, the reaction was initiated by adding 150 µL of HHL, and the mixture was incubated at 37 °C for 30 min. To terminate the reaction, 250 µL of a 1 mol/L HCl solution was introduced.

Following this, 2 mL of ethyl acetate was added to the mixture to extract hippuric acid (HA). The supernatant (1 mL) was evaporated to dryness using a Termovap Sample Concentrator after centrifugation at 10,000×g for 30 min. The residue was dissolved in 3 mL of ultrapure water, and the sample's absorbance was measured at 228 nm. The ACE inhibition rate was then calculated to assess the level of enzymatic inhibition in the experiment [17].

$$\text{ACE inhibition(\%)} = \frac{A_a - A_b}{A_a - A_c} \times 100\%$$

where A_a is the absorbance of HA generated without a sample (water instead of the sample), A_b is the absorbance of HA generated in the presence of sample, and A_c is the absorbance of HA generated in the absence of ACE which was inhibited enzymatic activity by initially adding HCl [17].

Cholesterol-Lowering Ability Evaluation

The tolerance to cholesterol and bile salts was tested according to the previous study with some modification [18]. EPS was weighed at various concentrations (0.4, 0.8, 1.2, 1.6 mg/mL) and mixed with 10-mL cholesterol standard solution, shaken and absorbed at 37 °C for 15 h, and then centrifuged at 10,000 r/min for 6 min. The supernatant was taken and applied by o-phthalaldehyde colorimetric method to determine the cholesterol concentration. Cholesterol standard solutions (1 mg/mL) were separated in various

volumes (0, 10, 20, 30, 50, 70, 90 µL) and then combined with glacial acetic acid to make a total volume of 400 L. Then, 2 mL of o-phthalaldehyde solution (0.2 mg/mL) was evenly distributed. Then, 1 mL of concentrated sulfuric acid was slowly added and mixed evenly. After 10 min at room temperature, the absorbance was measured at 550 nm, and a standard curve between cholesterol concentration and absorbance was created [19].

The cholesterol-lowering ability was calculated using formula.

$$\text{Cholesterol - lowering ability(\%)} = \frac{C_a - C_b}{C_a} \times 100\%$$

where C_a is the content of cholesterol generated without sample and C_b is the content of cholesterol generated in the presence of sample.

Phylogenetic and Pan-Genomics Analysis

The analysis of average nucleotide identity (ANI) on *L. plantarum* strains was performed to establish the taxonomic identification of the *L. plantarum* K25 strain (Table 1). After removing the plasmid, ANI analysis was performed using the JSpecies Web Server, which is publicly accessible at (<http://jspecies.ribohost.com/jspeciesws>). A clustering and heatmap were created using Heatmapper online tool (<http://www.heatmapper.ca/pairwise>). Furthermore, the pan-genome analysis was performed on Pan-Genome Explorer (<https://panexplorer.southgreen.fr>) (accessed on 10 August 2023) which is open-access online server for genomic analysis. Through a number of modules that make it easier to explore gene clusters and comprehend data, Pan-Genome Explorer do the pan-genome analysis based on the PGAP pipeline and presents the resulting information in a thorough and straightforward manner. The physical maps and synteny analysis were obtained as output [20].

Assessment for the Food Safety

The antibiotic resistance genes were analyzed using the CARD (Comprehensive Antibiotic Resistance Database) (<https://card.mcmaster.ca>), which has a significant number of annotated antibiotic resistance genes. Antibiotic resistance genes were discovered by comparing predicted genes to the CARD database with the use of particular algorithms and tools. On the detected matches, additional analysis and filtering were done based on factors such as sequence similarity, functional annotations, and resistance mechanisms. The found antibiotic resistance genes were classified and defined based on their resistance patterns [21]. Validation steps included antimicrobial susceptibility

Table 1 *Lactiplantibacillus plantarum* strains used in phylogenetic ANI and clustering analysis

Strains	Size	Contigs	GC %
<i>Lactiplantibacillus plantarum</i> K25	3,412,154	7	44.4
<i>Lactiplantibacillus plantarum</i> CGMCC 1.557	3,273,801	4	44.42
<i>Lactiplantibacillus plantarum</i> PS128	3,325,806	11	44.41
<i>Lactiplantibacillus plantarum</i> DSM 13273	3,439,800	90	44.22
<i>Lactiplantibacillus plantarum</i> WCFS1	3,348,624	4	44.42
<i>Lactiplantibacillus plantarum</i> UCMA 3037	3,108,278	68	44.52
<i>Lactiplantibacillus plantarum</i> NCU116	3,354,689	1	44.36
<i>Lactiplantibacillus plantarum</i> ST-III	3,307,936	2	44.48
<i>Lactiplantibacillus plantarum</i> JSA22	3,455,529	3	44.29
<i>Lactiplantibacillus plantarum</i> SRCM101060	3,263,056	101	44.33
<i>Lactiplantibacillus plantarum</i> DSM 20174	3,250,154	2	44.50
<i>Lactiplantibacillus plantarum</i> SRCM101060	3,263,056	101	44.33
<i>Lactiplantibacillus plantarum</i> 80	3,224,773	67	44.36
<i>Lactiplantibacillus plantarum</i> Nizo2877	3,231,771	2	44.50
<i>Lactiplantibacillus plantarum</i> WLPL04	3,185,263	51	44.52
<i>Lactiplantibacillus plantarum</i> B21	3,284,260	1	44.47
<i>Lactiplantibacillus plantarum</i> 8 RA-3	3,330,093	18	44.37
<i>Lactiplantibacillus plantarum</i> 90sk	3,371,458	47	44.26
<i>Lactiplantibacillus plantarum</i> WJL	3,477,495	102	44.24
<i>Lactiplantibacillus plantarum</i> IPLA88	3,254,055	208	44.43
<i>Lactiplantibacillus plantarum</i> ZJ316	3,299,755	4	44.49

testing and comparisons with previously reported resistance genes. This method allowed for the prediction and characterization of antibiotic resistance genes using the CARD database, providing light on the resistance profiles of bacterial isolates and expanding our understanding of the processes behind antibiotic resistance. To finalize the safety analyses of bacteria, VirulenceFinder 2.0 (<https://cge.food.dtu.dk/services/VirulenceFinder>) was used to estimate human host pathogenicity and identify acquired virulence genes [22, 23].

Prophage, Genomic Island, and Other Insertion Sequences Identification

The PHASTEST (PHAge Search Tool with Enhanced Sequence Translation) (<http://phastest.ca>) was used to identify prophage areas. It predicted prophage length, location, GC content, and gene annotation [24]. To validate the projected intact prophages, they were compared to the Virus-Host DB database (<https://www.genome.jp/virushostdb>). A proteomics tree of the viral genome sequence was also created using the VIPtree program. The tree was built using genome-wide sequence similarities estimated by tBLASTx [25]. This study contributed to a better understanding of the evolutionary links between viral genomes. Additionally, the existence of genomic islands (GI) was investigated using IslandViewer

4 (<https://www.pathogenomics.sfu.ca/islandviewer>). This enabled the identification and characterization of genomic islands in the bacterial genome. A detailed investigation of prophage areas and genomic islands was performed using these methodologies and databases, revealing useful insights into the genetic components present in the bacterial genome and their potential effect on bacterial evolution and virulence [26].

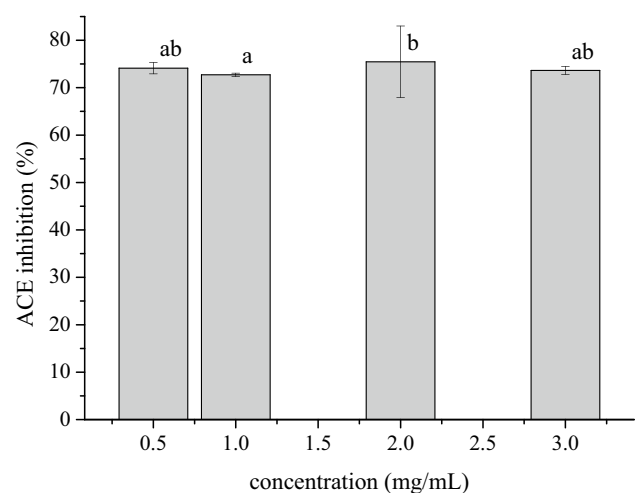


Fig. 1 Inhibition of angiotensin I-converting enzyme (ACE) enzyme by *L. plantarum* K25 EPS

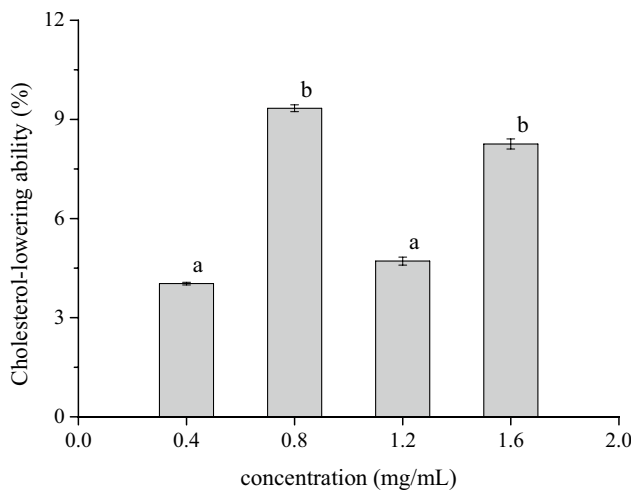


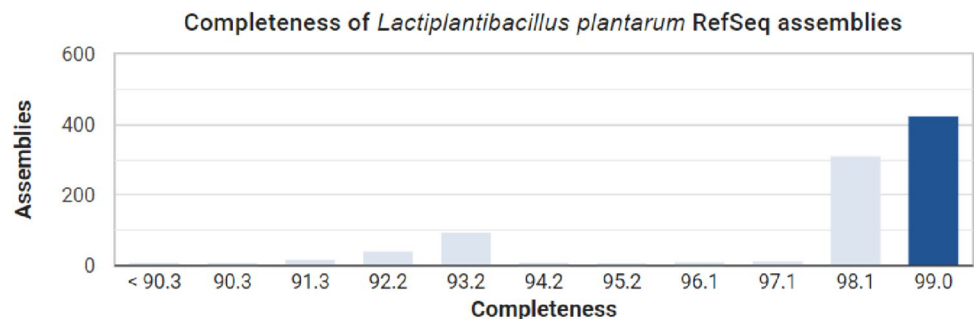
Fig. 2 Cholesterol-lowering activity of *L. plantarum* K25

Results

Angiotensin I-Converting Enzyme (ACE) Inhibitory Activity of K25 EPS

The o-phthalaldehyde colorimetric technique was used to evaluate the ACE inhibitory activity of K25 EPS. K25 EPS was tested at different doses (0.5, 1, 2, and 3 mg/mL), and the amount of hippuric acid (HA) that was produced was measured by taking absorbance readings at 228 nm. The computed ACE inhibition rates revealed a concentration-dependent inhibitory action, with values ranging from 74.5% at 0.5 mg/mL of *L. plantarum* K25 EPS to 77.0% at 3 mg/mL. These findings demonstrate *L. plantarum* K25 EPS's potential as an ACE inhibitor and point to its use in the control of hypertension and cardiovascular health. It is necessary to conduct more research on the mechanisms and pharmacological uses of K25 EPS in ACE inhibition. The graph given as Fig. 1 indicates the ACE inhibitory activity results as discussed above.

Fig. 3 The genome completeness overview of the genomic assemblies from *L. plantarum* K25



Cholesterol-Lowering Ability Evaluation

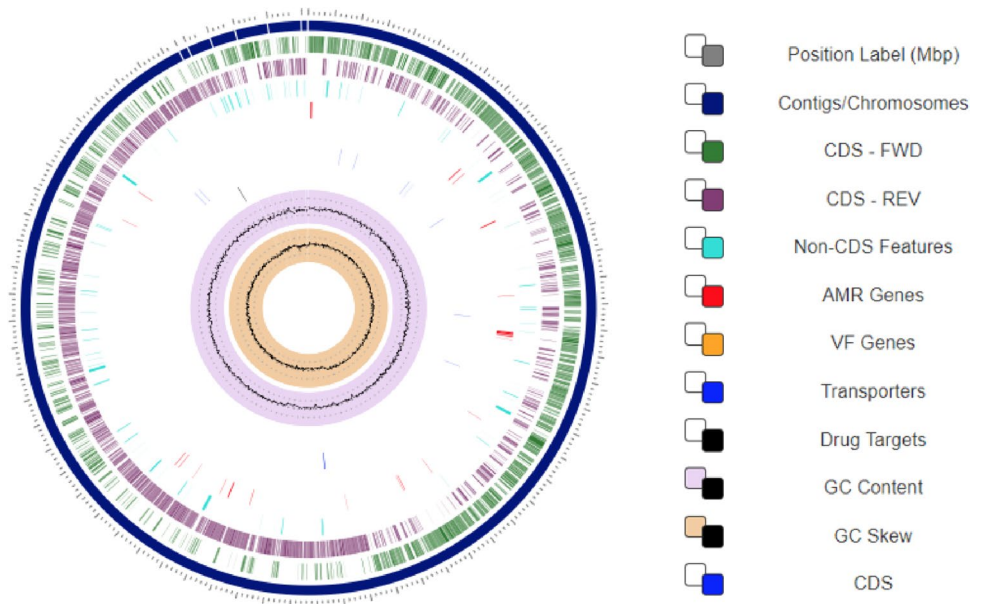
The EPS was mixed with a cholesterol standard solution and incubated at 37 °C for 15 h to test its capacity to decrease cholesterol at different concentrations (0.4, 0.8, 1.2, and 1.6 mg/mL). The supernatant was centrifuged, and the content of cholesterol was determined using an o-phthalaldehyde colorimetric examination of the supernatant. Using cholesterol standard solutions, a standard curve connecting cholesterol concentration to absorbance at 550 nm was produced. The capacity to decrease cholesterol was estimated using formula [19], where Ca stands for the cholesterol content produced without the sample and Cb stands for the cholesterol content produced with the sample present. The findings shed insight on the EPS's potential as a cholesterol-lowering drug by showing that it has diverse degrees of cholesterol-lowering efficacy at various doses. The cholesterol-lowering percentage was observed ranging from 3.3 to 9.2% at various concentrations of 0.0 to 0.2 (mg/mL). The graph given as Fig. 2 depicts the results of cholesterol-lowering activity.

Genomic Landscapes

The presence of mobile elements and repetitive regions within the genome of *L. plantarum* K25 poses difficulties in achieving complete genome closure using solely short-read sequences from the Illumina platform. The entire genome of *L. plantarum*, spanning 3,175,846 base pairs, exhibited a G + C content of 44.5% and was fragmented into 7 contigs. A thorough assessment of genome quality indicated a 99.35% level of genome completeness, accompanied by a minimal contamination rate of 1.72%. The completeness of the *L. plantarum* K25 RefSeq assembly is visually represented in Fig. 3. Figure 4 illustrates the genomic characteristics and circular visualization of *L. plantarum* K25.

By employing MetaGeneMark, the genes within the microbial genome of *L. plantarum* K25 were successfully predicted, resulting in the identification of a total of 3199

Fig. 4 The genetic features and circular perception of the *L. plantarum* K25



genes. These predicted genes exhibited characteristic lengths spanning from 200 to 3200 base pairs. Notably, the identification of expected genes was accompanied by an average intergenic spacing of 1000 bp, signifying a phenomenon of gene dispersion within the genome.

The exploration of the genomes of the studied organisms unveiled a diverse array of tRNA, tmRNA, and rRNA sequences. Within the tRNA category, distinct variations such as tRNA-Val, tRNA-Met, and tRNA-Ala were observed. Comparative analysis of the anticipated tRNA secondary structures highlighted disparities in both the anticodon loops and stems. Alongside tRNA, the presence of transfer-messenger RNA (tmRNA) was also anticipated in the genomes under investigation. This ribonucleic acid plays a pivotal role in facilitating the degradation of aberrant proteins during translation, thus serving as a vital component in the process of protein quality control. By envisaging the presence of tmRNA, a deeper comprehension of the mechanisms underpinning protein quality control mechanisms can be attained.

Phylogenetic and Comparative Analysis

ANI analysis was used to corroborate the taxonomy and phylogenetic relationship of *L. plantarum* K25, which was based on its whole genome assembly. Recent studies have shown that ANI analysis is a reliable tool for determining genetic and evolutionary divergence. This is because ANI analysis is based on a large number of conserved genes. As a result, multiple strains of *Lactobacillus plantarum* were included in the ANI analysis. The ANI analysis revealed that LPG1 had a high degree of similarity with other *L. plantarum* strains, with an ANI value above the recognized species identity

criteria of >95%. It was also discovered that *L. plantarum*. Figure 5 shows the clustering and heatmaps of distinct *L. plantarum* strains.

According to the insights derived from this study, the pan-genome is formed by an abundance of genes, presenting a remarkable scope and diversity. The study revealed a classification of genomes into distinctive categories, encompassing a set of genes universally prevalent across all living entities, alongside another set of genes exclusively shared among a specific subset of organisms. The differences noted in the sizes of supplementary genomes among prokaryotic organisms signify disparities in their genetic composition and paths of evolution. The functional interpretation of the pan-genome has revealed numerous pathways engaged in a range of biological functions, spanning metabolism, transportation, and regulation. Notably, variations were identified in the dispersion of genes linked to both virulence and antibiotic resistance across the studied species [27]. The distribution of COG functional categories obtained by pan-genome analysis is given in Fig. 6.

The incorporation of the pan-genome in constructing a phylogenetic strain tree has yielded valuable insights into the evolutionary interrelationships among these organisms. The outcomes derived from the strain tree analysis reveal distinct branches, serving as clear representations of genomic diversity and evolutionary heritage. Our study's findings propose that, for the most part, the taxonomic classification of these organisms concurs with their evolutionary links, albeit with only a few noteworthy exceptions. The observed distribution of gene counts as strain-specific genes was 30.3%, core genes as 52.8%, and dispensable genes 16.9%. The distribution of core genome vs accessory genome is given in Fig. 7, and the presence/absence matrix of hierarchical clustering is given in Fig. 8 [28].

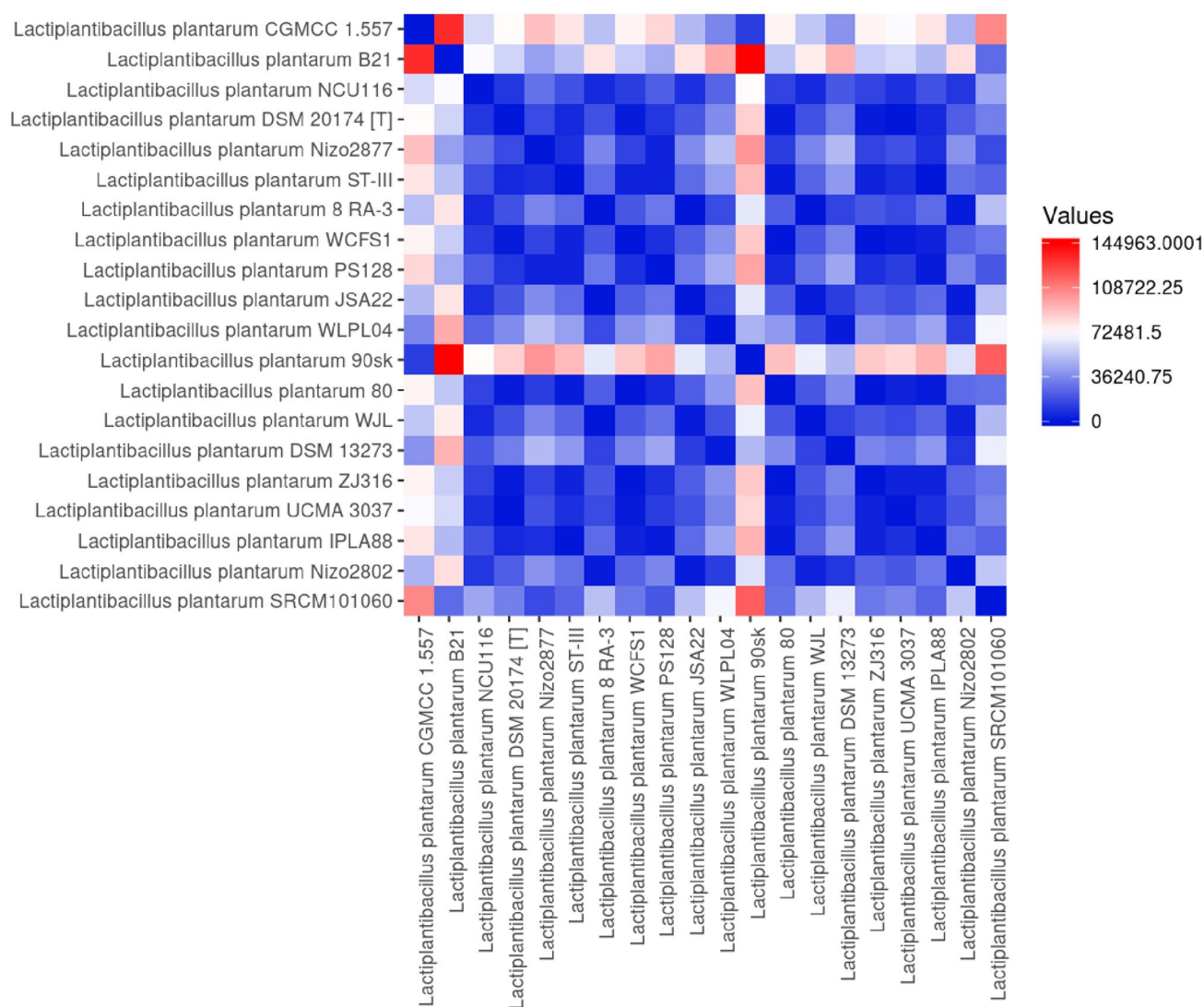


Fig. 5 Clustering and heatmaps of different strains of *L. plantarum* displaying the ANI percentage, aligned percentage, aligned bp, and total bp

Food Safety Assessment

A complete assessment of food safety for the *L. plantarum* K25 genome was performed using the Complete Antibiotic Resistance Database (CARD). We systematically analyzed the existence of antibiotic resistance genes within *L. plantarum* K25 using CARD analysis. The results of this investigation indicated that the strain lacks any known antibiotic resistance genes, indicating a good antibiotic resistance profile. However, two genes, VanY and qacJ, were found with a low score in the genome, signaling a possible risk for vancomycin and ammonium quaternary disinfectants resistance.

The finding of antibiotic-resistance genes among probiotic strains might reasonably arouse worries about the

spread of resistance to dangerous bacteria, stressing the serious consequences for food safety. This study's use of CARD gave significant insights by demonstrating the lack of antibiotic resistance genes. This discovery significantly strengthens *L. plantarum* K25's safety profile, placing it as a suitable option for probiotic applications. These findings not only add to our understanding of the strain's appropriateness for inclusion in food items but also highlight its potential as a safe and beneficial probiotic component [21]. Furthermore, no matches were detected using VirulenceFinder 2.0, showing that it is safe and non-pathogenic for human ingestion [29]. Figure 9 below depicts the antibiotic resistance wheel obtained from CARD Database.

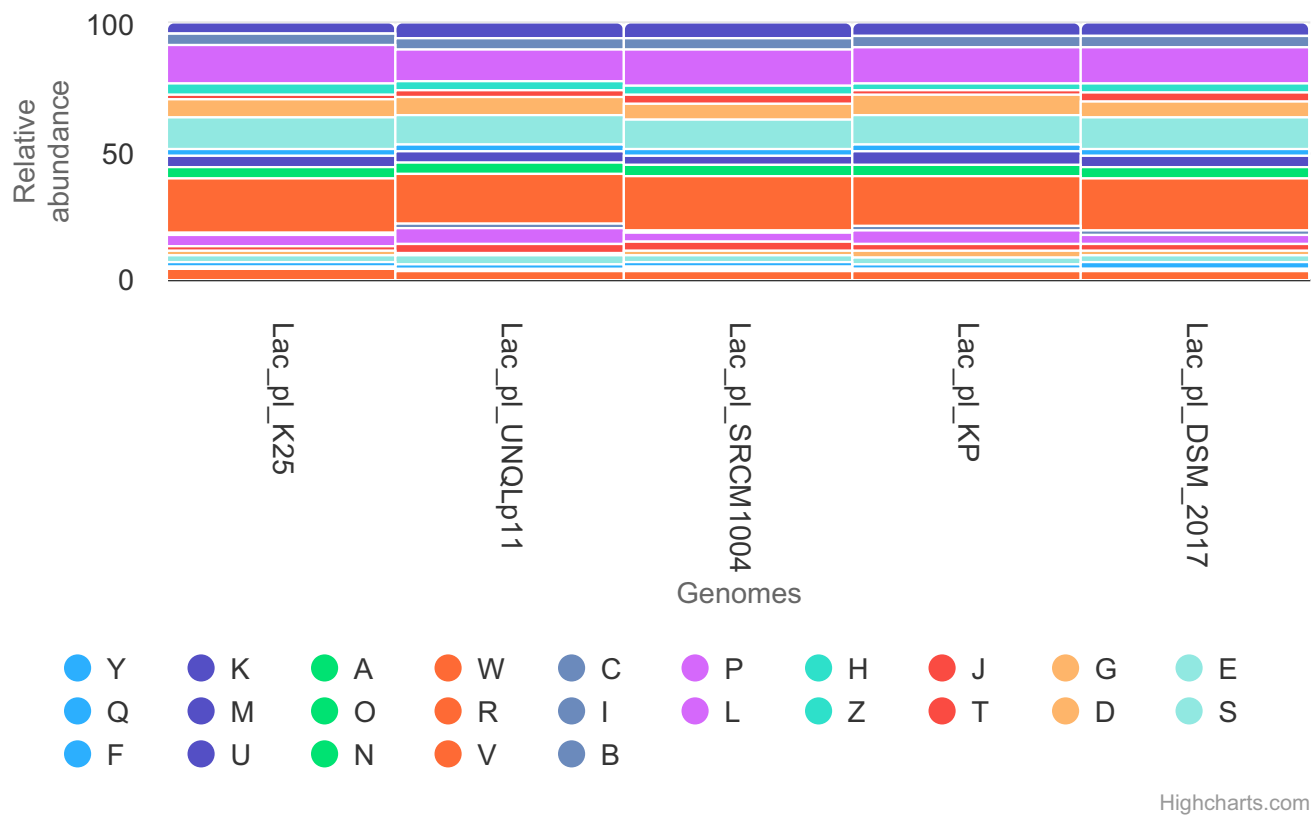


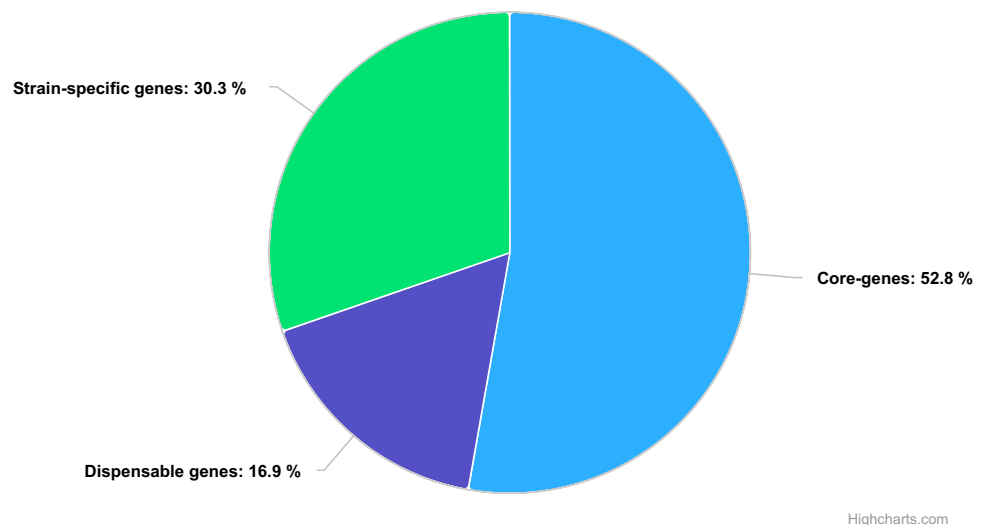
Fig. 6 Distribution of COG functional categories defined by Pan-Genome Explorer

Prophage, Genomic Island, and Other Insertion Sequences Identification

The PHASTEST (PHAge Search Tool with Enhanced Sequence Translation) (<http://phastest.ca>) was used to locate and describe prophage areas within the genomic material. This powerful technology accurately predicted

the length, location, GC content, and gene annotation of these prophages. In order to validate the accuracy of the predicted intact prophages, the regions identified were cross-referenced with the Virus-Host DB (<https://www.genome.jp/virushostdb>) database. This comparative method not only confirmed the presence of the prophages but also revealed their possible host range. As a consequence, 4 different

Fig. 7 Distribution of core genome and accessory genome in *L. plantarum* K25



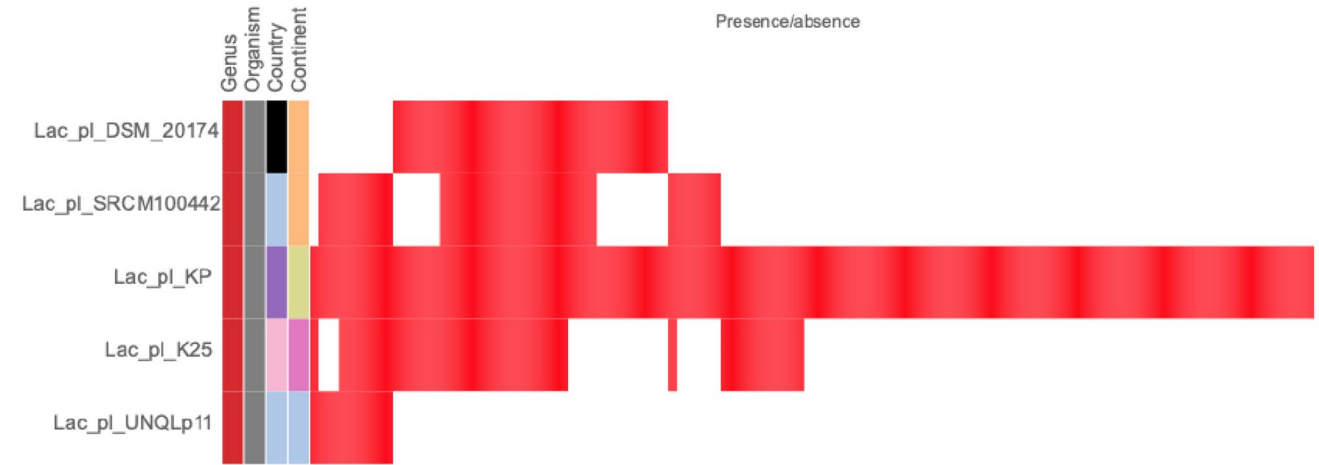


Fig.8 The presence/absence matrix based on gene clusters ordered using a hierarchical clustering

phage regions were effectively identified along with the 129 phage genes and 176 bacterial genes, as shown in Table 2, while Fig. 10 depicts a circular genome map with prophage areas highlighted.

The VIPtree tool was also used to create the proteome tree of the viral genomic sequence. Based on genome-wide sequence similarities, this tree assisted in visualizing the evolutionary links across viral genomes. A comprehensive

Fig. 9 Antibiotic resistance gene families in *L. plantarum* K25 are depicted on an antibiotic resistance wheel

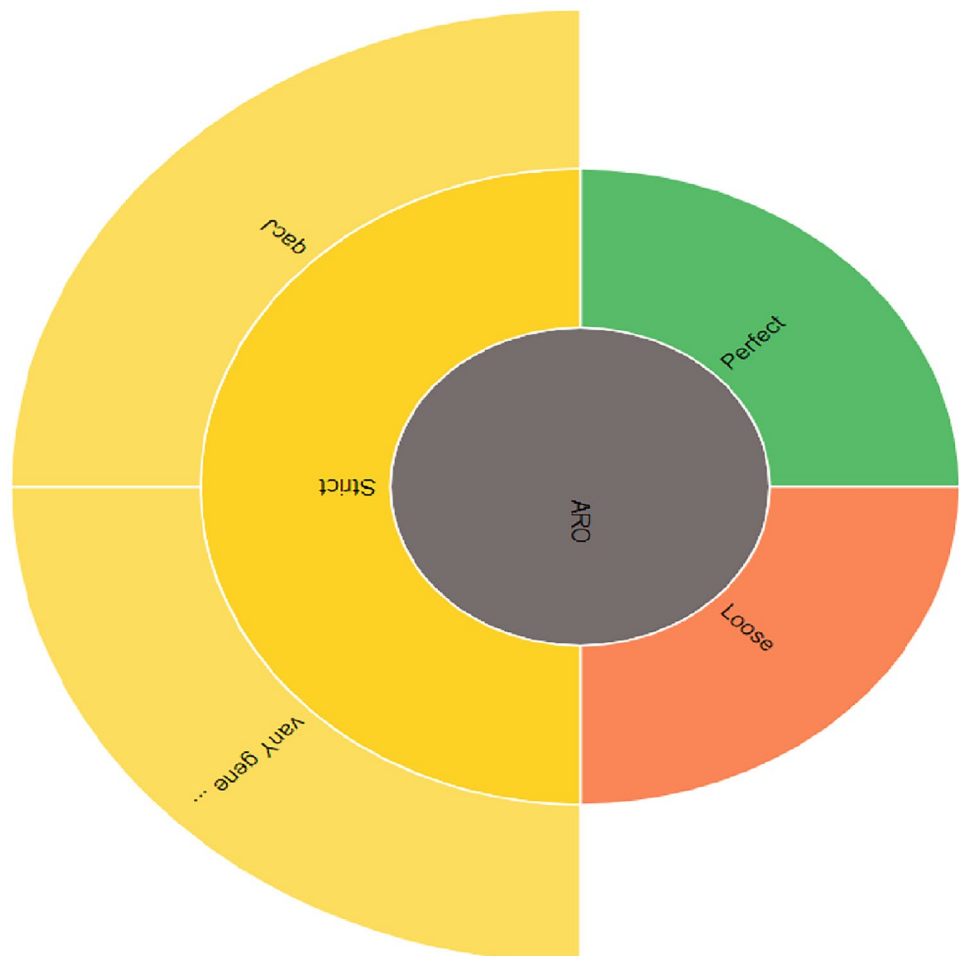


Table 2 Predicted prophage region, their length, localization, and GC content by PHASTEST

Region	Length	Score	Proteins	Position	Most common phage	GC%
1	16.9 Kb	70	25	36009–52937	PHAGE_Strept_315.2_NC_004585(3)	42.06%
2	45.4 Kb	150	18	572363–617,839	PHAGE_Lactob_phig1e_NC_004305(7)	40.81%
3	39.4 Kb	150	40	1798510–1837935	PHAGE_Lactob_Sha1_NC_019489(19)	40.78%
4	42 Kb	80	9	2111511–2153533	PHAGE_Lactob_Sha1_NC_019489(3)	43.38%

analysis of the prophage regions was performed using the PHASTEST tool, database comparisons, and the VIPtree application, enhancing our understanding of the genetic elements present in the studied genomes and their potential impact on bacterial evolution and host–pathogen interactions. Figure 11 depicts the circular proteome tree of the viral genomic sequences.

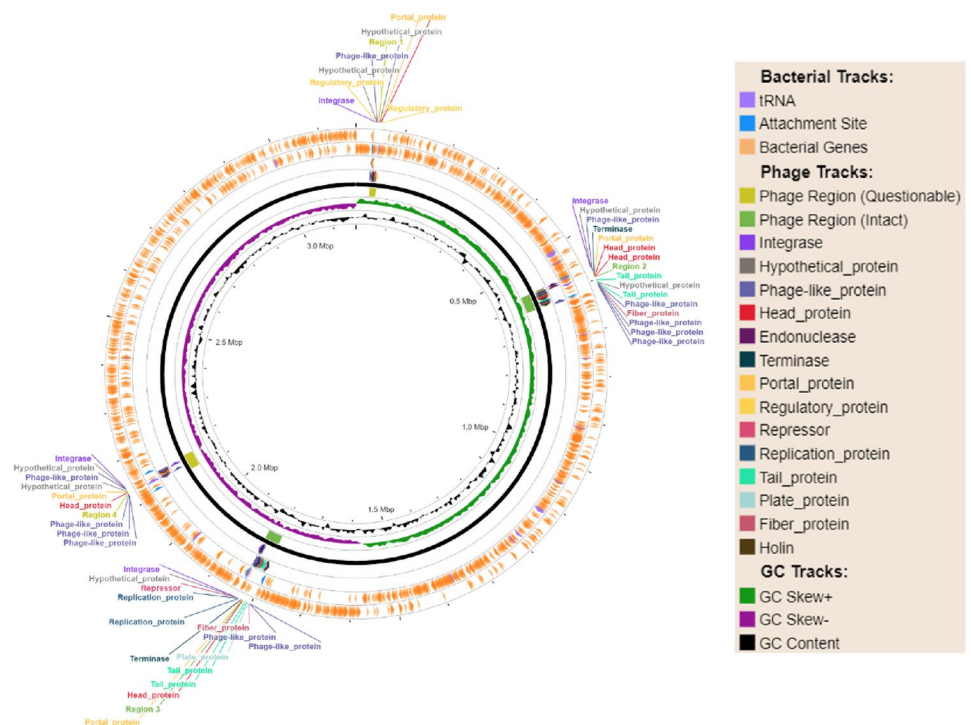
The bacterial genome was meticulously examined to determine the presence of genomic islands (GIs) using the capabilities of the IslandViewer 4 program. Several databases were used in this work to identify and define these genomic islands within the genome. The discovery of genomic islands was made possible by this thorough investigation, revealing light on both their structural and functional characteristics. A comprehensive understanding of the existence and properties of genomic islands in the bacterial genome was attained by utilizing the capabilities of IslandViewer 4 and carefully studying multiple databases. These results increase our understanding of the genetic composition of the bacterium under study and shed light on the potential functions of genomic islands in bacterial evolution,

adaptability, and virulence. Using IslandViewer 4, a total of 40 genomic islands were predicted. Table 3 provides other information, such as size and GI prediction techniques.

The circular view of the genomic islands is given in Fig. 12, depicting the highlighted regions for the genomic islands.

Notably, K25 EPS exhibits strong angiotensin I-converting enzyme (ACE) inhibitory action, implying a possible involvement in hypertension and cardiovascular health. Furthermore, K25 EPS has varying levels of cholesterol-lowering effectiveness, indicating its potential as a functional dietary ingredient. Genomic research revealed the presence of mobile elements and prophage regions in *L. plantarum* K25, providing insight into genetic diversity and evolution. Importantly, the absence of antibiotic resistance genes enhances its safety profile for probiotic uses, making it suitable for food and health goods. This multidimensional study sheds light on the health-promoting features and genetic characteristics of *L. plantarum* K25, paving the door for future research in functional food and pharmaceutical industries.

Fig. 10 The anticipated prophage regions in the *L. plantarum* K25 genome are shown in a circular genomic perspective



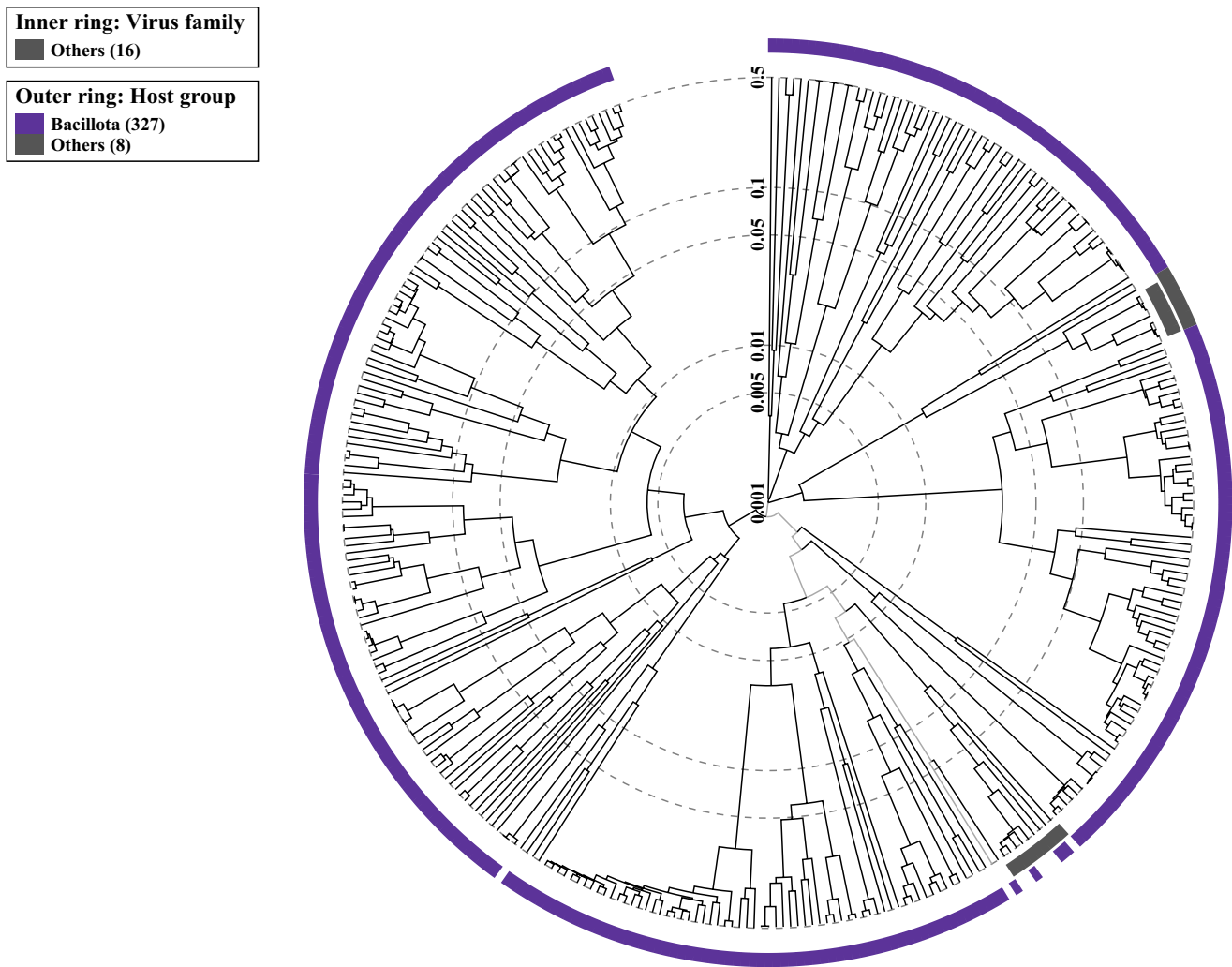


Fig. 11 Circular proteomic tree of the viral genomes of the innermost ring symbolizes the virus family of phages, while the outermost ring represents the host group of bacteriophages

Discussion

The present study explores the *L. plantarum* K25 genome, displaying complex intricacies backed by solid data. The genome's 3.2 million base pairs were broken up into 7 contigs, boasting great completeness and low contamination, despite difficulties in genome closure caused by repetitive regions. A total of 3199 genes with interesting patterns of dispersion were discovered throughout the gene identification procedure. Understanding translation mechanisms was made possible by studying the tRNA, tmRNA, and rRNA sequences [30]. The genetic similarity of *L. plantarum* K25 to other strains was confirmed by phylogenetic analysis. The strain tree showed evolutionary linkages, while the pan-genome analysis revealed various genes grouped into general and particular groupings. The investigation demonstrated the lack of pathogens and antibiotic resistance genes,

which was further substantiated by the detection of prophage and genomic islands. This thorough investigation advances knowledge of the complex genetic features and possible uses of *L. plantarum* K25 [5].

When the current study is compared to earlier studies, a number of noticeable discrepancies and surprising conclusions are revealed. First off, the examination of the *L. plantarum* K25 genome in the present work is unique in its thorough methodology, offering insights into both its genetic makeup and possible uses [3]. In contrast, earlier research could have concentrated just on a few factors, missing the whole genetic study shown here. The surprising discovery of 3199 genes with fascinating dispersion patterns also calls into question accepted wisdom and points to a distinct genetic structure within *L. plantarum* K25. This is a change from earlier research that could have presupposed more uniform patterns of gene distribution [31].

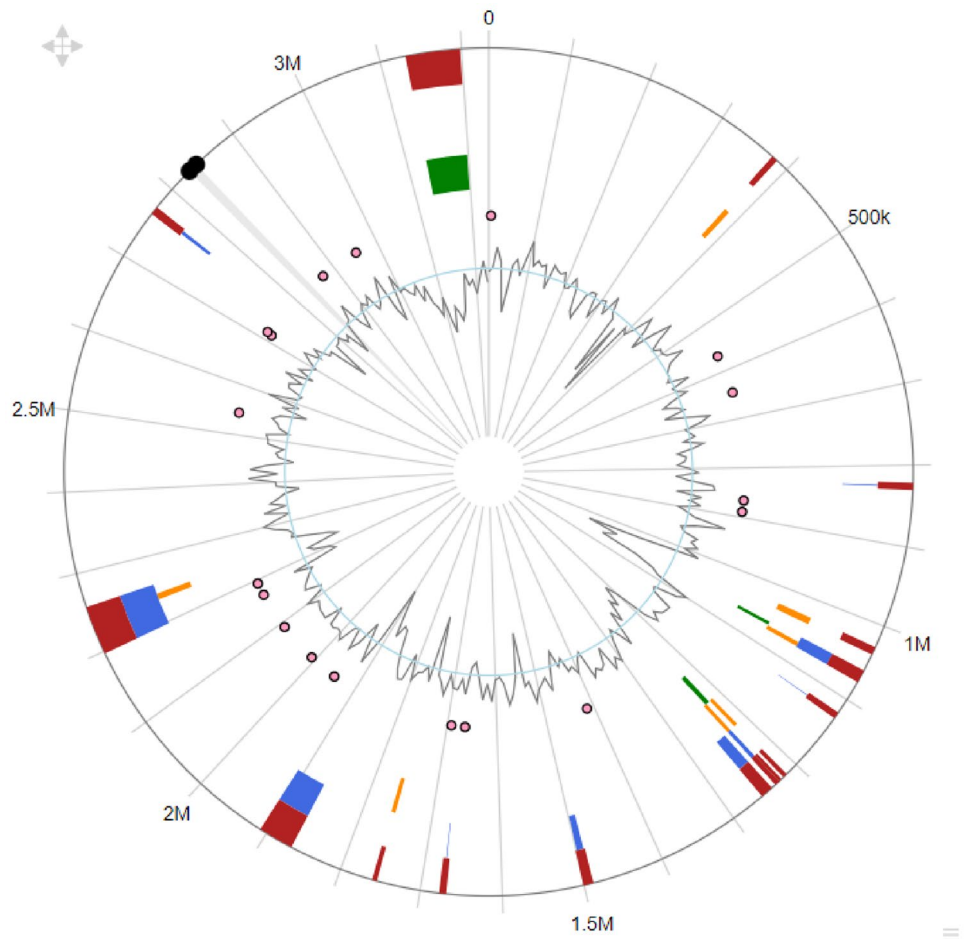
Table 3 The predicted genomic islands, their sizes, and prediction methods

Start	End	Size	GI prediction method
376,462	384,107	7645	Predicted by at least one method
821,356	830,144	8788	Predicted by at least one method
1,027,423	1,037,372	9949	Predicted by at least one method
1,058,893	1,074,987	16,094	Predicted by at least one method
1,060,726	1,066,039	5313	Predicted by at least one method
1,067,018	1,073,422	6404	Predicted by at least one method
1,118,943	1,127,380	8437	Predicted by at least one method
1,216,293	1,221,709	5416	Predicted by at least one method
1,225,029	1,229,221	4192	Predicted by at least one method
1,229,402	1,234,901	5499	Predicted by at least one method
1,229,402	1,234,901	5499	Predicted by at least one method
1,240,894	1,256,538	15,644	Predicted by at least one method
1,488,171	1,500,777	12,606	Predicted by at least one method
1,668,295	1,676,832	8537	Predicted by at least one method
1,753,863	1,759,812	5949	Predicted by at least one method
1,865,399	1,908,848	43,449	Predicted by at least one method
2,197,131	2,257,782	60,651	Predicted by at least one method
2,234,614	2,243,732	9118	Predicted by at least one method
2,761,319	2,771,572	10,253	Predicted by at least one method
3,131,242	3,198,242	67,000	Predicted by at least one method
1,060,726	1,066,039	5313	IslandPick
1,225,029	1,229,221	4192	IslandPick
3,131,242	3,198,242	67,000	IslandPick
376,462	384,107	7645	SIGI-HMM
1,027,423	1,037,372	9949	SIGI-HMM
1,067,018	1,073,422	6404	SIGI-HMM
1,216,293	1,221,709	5416	SIGI-HMM
1,229,402	1,234,901	5499	SIGI-HMM
1,753,863	1,759,812	5949	SIGI-HMM
2,234,614	2,243,732	9118	SIGI-HMM
821,356	830,144	8788	IslandPath-DIMOB
1,058,893	1,074,987	16,094	IslandPath-DIMOB
1,118,943	1,127,380	8437	IslandPath-DIMOB
1,229,402	1,234,901	5499	IslandPath-DIMOB
1,240,894	1,256,538	15,644	IslandPath-DIMOB
1,488,171	1,500,777	12,606	IslandPath-DIMOB
1,668,295	1,676,832	8537	IslandPath-DIMOB
1,865,399	1,908,848	43,449	IslandPath-DIMOB
2,197,131	2,257,782	60,651	IslandPath-DIMOB
2,761,319	2,771,572	10,253	IslandPath-DIMOB

Furthermore, the absence of antibiotic resistance genes in *Lactiplantibacillus plantarum* K25 is a noteworthy finding that differs from several past studies that could have discovered such genes. This surprising result highlights the strain's potential safety and usefulness as a probiotic ingredient. The discovery of prophage and genomic islands, together with their possible influence on evolution and virulence, gives the research

a fresh perspective and distinguishes it from earlier studies that might not have thoroughly investigated these genetic components [32]. This study overcomes obstacles to show the genetic diversity, composition, and safety characteristics of *L. plantarum* K25. The results of the study, including phylogenetic links and the lack of genes for antibiotic resistance, highlight the strain's potential for use in a variety of applications.

Fig. 12 Circular map of the genomic islands predicted by IslandViewer 4



Conclusions

The analysis of the *L. plantarum* K25 genome has revealed information on its genetic makeup and possible uses. A better knowledge of the genetic makeup of this bacterial strain has been made possible by the study's all-encompassing methodology, which ranged from genome completion to gene identification and functional annotation. Unexpected results that challenge preconceived notions and emphasize the distinctiveness of *L. plantarum* K25 include the different gene dispersion patterns and the absence of genes associated with antibiotic resistance. Additionally, the study's extensive analysis of genomic and prophage islands has improved our knowledge of potential evolutionary implications. Together, these findings highlight the strain's suitability for probiotic usage and its potential advantages in a range of contexts. In the end, this study broadens our understanding of the intricate genetic makeup of *L. plantarum* K25 and highlights its potential to progress in several areas, including biotechnological applications and food safety.

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Availability of Data and Materials All the data produced or generated during the study has been given in the manuscript.

Declarations

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Conflict of Interest The authors declare no competing interests.

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